

TAKUYA BOEHRINGER

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MS Mechanical Engineering student at Columbia University. Specialized in robotics and autonomous systems. British and Canadian.

EDUCATION

MS Mechanical Engineering (Robotics and Control Track) - Columbia University	August 2025 - Present
MEng Computer Science (Robotics Minor) - University College London (UCL)	Sep 2021 - June 2025

PUBLICATIONS

[Conference Paper] IEEE International Conference on Automation Science and Engineering (CASE)

First Author

Immersive Teleoperation Framework for Locomanipulation Tasks

[Journal Paper] Scientific Reports (Nature)

Second Author

Analyzing Spatio-temporal Dynamics of Dissolved Oxygen for the River Thames using Superstatistical Methods and Machine Learning

WORK EXPERIENCE

Robotics Engineering Intern - Animo Robotics - (Perception) May - July 2026

- Built the **perception stack** for a robotic cell designed for food prep.
- Filtered the output pointcloud to get a reliable scooping point, calculated in robot frame without discretizing it.
- Developed the **calibration procedure** and worked within spacial accuracy, precision, runtime and compute requirements.
- Made an extensive **testing harness** to validate depth accuracy under various conditions, analyzing the results afterwards.
- Also used a **neural segmentation model** to localize the conveyor belt running past the robot in real time.
- Ran an accuracy study to compare cameras, and made the camera an interchangeable component for future improvements.

Summer Researcher - UCL - (Robot Perception and Learning Lab) May - Sept 2024

- Collaborated with **Prof Dimitrios Kanoulas** to have my paper accepted by the **IEEE International Conference on Automation Science and Engineering (CASE)**.
- Presented the paper live at **CASE 2025** in **Los Angeles**.
- Developed a framework for VR robotic teleoperation for locomanipulation tasks.
- Used **Unity** to develop the interface for the **Meta Quest 2**, integrating **ROS2** and **Gaussian Splatting** into the pipeline.

Software Engineering Intern - Leonardo - (Electronic Warfare) July - Aug 2023

- Worked on a classified project in the **Electronic Warfare** division.
- Used C++ for **embedded linux programming** and to interface with firmware.
- Programmed in C# to create a **WPF user interface** and **socket programming** to interact with embedded devices.

PROJECTS

Columbia Robotic Manipulation and Mobility Lab (ROAM) Feb - May 2026

- Worked on the **MiniBEE** project, a bimanual robotic manipulator using two reduced-mobility arms coupled into a kinematic chain.
- Working on automated kinematic optimization under **Prof Matei Ciocarlie**.

Autonomous Robotic-Arm Docking System for Spacecraft Jan 2026 - May 2026

- Designed an RPR (revolute-prismatic-revolute) robotic manipulator for **spacecraft docking**.
- Built a small-scale prototype for the Intro to Human Spaceflight class taught by former astronaut **Prof Mike Massimino**.
- Used **Onshape** for the CAD and simulated it in **Isaac Sim**.

Robot Studio Project Sept - Dec 2025

- Designed, built and programmed a robot of my own design for the Robotic Studio class by **Prof Hod Lipson**.
- Built two **Klann linkages** for an eight-legged locomotion.
- Trained it to crab walk via simulation in **MuJoCo** using various optimization algorithms and **reinforcement learning**.
- Deployed the policy **sim-to-real** for testing.

SKILLS

Programming Languages: *Python, C, C++, C#, MATLAB*

Machine Learning: *Supervised and Unsupervised Learning, Deep Learning (PyTorch), Reinforcement Learning*

Electronics: *Analog Electronics, Microcontrollers (Arduino), Schematics (KiCAD)*

Simulation: *MuJoCo, Isaac Sim, Unity, RoboDK*

CAD/CAM - *Fusion 360, Onshape*

Other: *ROS2, Git, Docker, Linux, 3D Printing*